A pre-registered replication of the Sawin–Sutherland height-ordered murmuration

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Methods disclosure. The research design, the mathematical decisions, the pre-registrations, and all review are the author's. The C++ implementation, the documentation drafts, and the execution of the analyses were done by an AI assistant (Anthropic's Claude) under the author's direction, following the repository's working rules: dual independent implementations, oracles as referees only, decision rules committed before the data they judge, no constant taken from memory. The author verified the artifacts and is responsible for this record.

Abstract

We re-derive the Sawin–Sutherland height-ordered elliptic-curve murmuration from scratch and probe it against their conjectured Bessel-J_1 density over a four-rung height ladder, $X = 10^4$ ($\approx 2^{13.3}$), 2^{16}, 2^{17}, 2^{18}. An initial comparison showed an apparent persistent trough displacement of ≈ 0.0825 against a density target of 0.805. A pre-registered audit round traced that displacement to a transcription error in our reading of the density's defining equation: the two local-factor products were transposed and one exponent mis-set, which had mislocated the conjectured trough. We log the correction as erratum #28. Corrected, the three shape invariants fixed in advance (the positive hump, the first post-hump zero, and the trough) agree with the density within the a-priori tolerance $\tau = 0.06$, the largest residual ≈ 0.012 at the trough. Two pre-registered follow-ups bound candidate mechanisms: excising the analytic-rank-2 subpopulation does not move the trough, and root-number-imbalance leakage is $\leq 0.5\%$ at the reliable rungs. Every result is numerical over a stated finite range, twinned or oracle-refereed, and reproducible across compilers and platforms; we claim replication and constraint, not proof.

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1. Introduction

Murmurations are an oscillating correlation between the Frobenius traces $a_p(E)$ of a family of elliptic curves and the prime p. He–Lee–Oliver–Pozdnyakov discovered them in 2022; the phenomenon has since been observed in several arithmetic settings, and proved in the first of them, Zubrilina's weight-2 density. The companion explainer (E1) gives a fuller account.

Sawin–Sutherland [SS25] order the family by naive height rather than by conductor and propose an explicit candidate for its limiting density. Two of their statements must be kept apart; the distinction fixes what this note claims. The first is a conjecture, the second a theorem.

- **Conjecture 1 (the density).** For $0 < C_1 < C_2$, [SS25] conjecture that the $X \to \infty$ limit of their statistic (1) equals $\int_{C_1}^{C_2} D(u)\,du$, where the murmuration density is the Bessel-J_1 double sum of their equation (2):

$$D(u) = 2\pi\sqrt{u} \sum_{\substack{q\in\mathbb{N} \\ \text{squarefree}}} \sum_{m\in\mathbb{N}} \frac{\mu(\gcd(m,q))}{q\,m\,\varphi(q/\gcd(m,q))} \times J_1(4\pi\sqrt{u}\,m/q) \prod_{p\mid q} \hat{\ell}_{p,2v_p(m)} \prod_{\substack{p\mid m \\ p\nmid q}} \ell_{p,2v_p(m)}, \tag{2}$$

where $u = p/N$. Here J_1 is the Bessel function of the first kind, v_p the p-adic valuation, μ the Möbius function, and φ Euler's totient; the local factors $\ell_{p,\nu}$ (their Lemma 3, positive ν) and $\hat{\ell}_{p,\nu}$ (their Lemma 4, even $\nu \geq 0$) carry the paired products — $\hat{\ell}$ over $p \mid q$ and ℓ over $p \mid m$, $p \nmid q$, both at exponent $2v_p(m)$. An earlier reading of this line transposed the two products and mis-set the $p \mid q$ exponent; the correction (erratum #28) is the subject of §6.

- **Theorem 2 (the proven variant).** [SS25] prove a variant in which the prime sum is replaced by a sum over integers with no prime factor $\leq P$, against a smooth weight; the same integrand appears, and the $P \to \infty$ limit is established. This smoothed variant motivates Conjecture 1 without being the prime-sum murmuration.

The explicit density of the height-ordered prime-sum murmuration is therefore conjectural. The proven Theorem 2 is a rigorous backdrop, not the object our empirical curve is compared against. This note replicates and probes Conjecture 1; it proves nothing. Throughout, "verified" means numerically, over a stated finite range, cross-checked by two independent implementations, by a published anchor value, or by an external oracle. It never means a formal proof.

What this note tests. We compute statistic (1) from scratch over the height family up to $X = 10^4, 2^{16}, 2^{17}, 2^{18}$, bin it by $u = p/N(E)$, and compare the empirical density to the conjectured $D(u)$ through three shape invariants fixed in advance: the positive hump, the first zero crossing after it, and the trough (§3). Two pre-registered follow-ups then bound candidate mechanisms for the residual departure of the empirical curve from $D(u)$ (§4).

What this note does not test. It makes no asymptotic ($X \to \infty$) claim. Our computable range, $X \leq 2^{18}$, sits at the bottom of the window over which [SS25] observe the bias and expect it to decay, naive height 2^{16} to 2^{28}, reached only through sub-linear point counting. Every finite-range statement below concerns that range, not a limit; the note makes no persistence claim (§3). The discussion (§6) gives a documented precedent in which a small-conductor picture does not survive to the asymptotic one; that is why the distinction is not pedantic.

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2. Methods

Family and ordering. We use the short Weierstrass family $E_{A,B} : y^2 = x^3 + Ax + B$ with integers A, B, *reduced* (no prime p with $p^4 \mid A$ and $p^6 \mid B$, the unique minimal representative of each isomorphism class) and *nonsingular* ($4A^3 + 27B^2 \neq 0$), ordered by the naive height $H(E) := \max(4|A|^3, 27|B|^2)$, quoted verbatim from [SS25] p.1. The family $\{E_{A,B} : H(E) \leq X\}$ is enumerated directly over the box $|A| \leq (X/4)^{1/3}$, $|B| \leq (X/27)^{1/2}$; its cardinality is certified by a second, independent sieve algorithm required to agree with the direct count at every cutoff.

The two objects, side by side. The note compares one *computed* quantity with one *conjectured* one. The distinction (§1) governs what the note claims.

Empirical: a binned estimator of [SS25] statistic (1). For a prime-ratio window $(C_1, C_2]$ (half-open, left-open and right-closed, as in [SS25] p.2 — see §5 and erratum #30), [SS25]'s statistic (1) is

$$\mathbb{E}_{\{E\,:\,H(E)\le X\}}\left[\frac{\log\big(N(E)^{(C_1+C_2)/2}\big)}{N(E)}\sum_{\substack{p/N(E)\in(C_1,C_2]\\ p\text{ prime}}}\epsilon(E)\,a_p(E)\right]. \tag{1}$$

The two displayed equations keep [SS25]'s numbers, (1) and (2); the binned estimator $\hat{D}$ below is ours and is left unnumbered.

We evaluate it as a density on $(0,1]$: partition into 40 bins of width $\Delta u = 0.025$, take each bin $b = (b\,\Delta u,\,(b+1)\Delta u]$ as the window (so its midpoint $(C_1+C_2)/2 = u_{\text{mid}} := (b+\frac{1}{2})\,\Delta u$), and divide by Δu:

$$\hat{D}(u_b) = \frac{1}{\Delta u\,|\text{fam}|}\sum_{(E,p)\,:\,p/N(E)\,\in\,\text{bin}_b}\left(u_{\text{mid}}\,\frac{\ln N(E)}{N(E)}\right)\epsilon(E)\,a_p(E),$$

over good primes $p > 3$ with $p \nmid (4A^3+27B^2)$ and $p \le N(E)$ (so $u \le 1$). The u_{mid} factor is eq (1)'s $(C_1+C_2)/2$ normalization exponent evaluated at the bin midpoint, not a per-pair weight, so $\hat{D}$ is a binned density estimator of statistic (1), one bin per window. For this family the reduced short model is p-minimal at every $p > 3$, so $p \nmid \Delta(\text{model}) \iff p \nmid N(E)$ (good reduction), and this exact good-prime test keeps a_p's domain computed-only.

Conjectured: [SS25] eq. (2) / Conjecture 1. [SS25] conjecture that the $X \to \infty$ limit of (1) equals $\int D(u)\,du$, where $D(u)$ is the Bessel-J_1 density transcribed verbatim in §1. eq (1) is a definition we evaluate; eq (2)'s equality to the limit is the *conjecture*. $\hat{D}$ is compared to D; the note reports empirical agreement, never a proof (§1).

Provenance: computed vs. oracle. The Frobenius trace $a_p = p + 1 - \#E(\mathbb{F}_p)$ is computed from scratch. The conductor $N(E)$ and root number $\epsilon(E)$ are oracle-provenance input supplied by PARI/GP, trusted input *to* the statistic and not a referee *of* our numbers. Every emitted column is labelled with its provenance. The oracle N/ϵ are certified before use by a dual-oracle overlap check against LMFDB/Cremona data.

Twin / anchor / oracle architecture. Each function the result depends on has two independent implementations (a different algorithm, never a refactor of itself) or an external oracle check. a_p is computed by a fast quadratic-residue-table character sum (`ap_fast`) twinned against a frozen modular-exponentiation referee (`ap_charsum`); the two agree exactly on a certified sample. Published values are pinned as anchor tests; oracles (PARI, LMFDB, Odlyzko) referee and are never called from the computational core; an absent oracle skips visibly rather than passing silently. The full method is described in the companion method explainer (E3).

a_p is computed by three independent algorithms: the frozen `ap_charsum` referee, the fast QR-table `ap_fast`, and Shanks–Mestre point counting. They agree exactly over the full 2^{16} and 2^{17} a_p grids (112 M and 385 M values, 0 mismatches), at about 145× less CPU aggregated over the 2^{16} grid ($O(p^{1/4})$ point counting vs. $O(p)$ character sum). At 2^{18} the equality is verified on a tail-weighted 27.9 M-value sample against the $O(p)$ referee and on a PARI `ellap` spot (204 pairs, p up to 6.9 M), the pre-registered production-capability gate. This is a cross-algorithm check (Shanks–Mestre vs. the charsum referee). Since a_p is a platform-independent integer, it is verified same-platform by design, not as a two-platform claim.

Pre-registration. Every decision rule, tolerance, and threshold is committed to version control *before* the data it judges exists, cited here as (pre-registered → read) commit pairs. PR-1, the ladder decision rule and finite-X/persistent fork: pre-registered `dd6beb0`, with the 2^{18} Rung-3 clause `4a17ebe` committed *before the* 2^{17} *rung was read*, read `8f64ba1` (verdict H0, $\le 2^{18}$). PR-2, the analytic-rank contrast threshold: pre-registered `f7415a4` *before any split curve*, read `876999f` (null). PR-3, the leakage estimator: pre-registered `21060a0`, read `0d21b62`. Every exploratory statistic looked at is recorded in a looks ledger.

The conjectured density. $D(u)$ is evaluated independently of the empirical statistic (a separate translation unit, authored without reference to the empirical side) as the Bessel-J_1 double sum of Conjecture 1 with the local factors of [SS25] Lemmas 3–4, an in-house J_1, and *generated* constants (never typed), truncated at $q, m \leq 2000$. Its reading of eq (2) is transcribed from the [SS25] PDF, not from an internal quote — the standing rule that a transcription check terminates at the source artifact, adopted after erratum #28 (§6). The local factors carry level-1 Hecke eigenvalue sums $\sum_f a_p(f) = \mathrm{Tr}(T_p \mid S_{\nu+2})$, which we compute by a from-scratch Eichler–Selberg trace reusing the M3 Hurwitz class numbers, so no oracle-provenance constant enters the density's core (rule 3); the trace is anchored on the published $\mathrm{Tr}(T_p \mid S_{12}) = \tau(p)$. The density is checked at two levels: term-level agreement on hand-picked structural survivors (the $m = 32$ term through $\mathrm{Tr}\, T_2$, the $p = 3$ branch through $m = 3^5$, and $\gcd(m, q) > 1$ cases exercising the Möbius factor), and curve-level agreement of the whole density. The C++ evaluator and an independent PARI/GP oracle built from the same Lemmas are two codings of one reading, so their agreement alone does not certify that reading; the external referee is the authors' published figure, whose trough the corrected density matches at $u \approx 0.87$ — the reconciliation recorded as erratum #28 (§6). Because every value is built from IEEE operations, generated constants, and the in-house Bessel, the emitted curve is byte-portable across compilers and platforms, a property we verify (§5).

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3. Results

The ladder. Over four height cutoffs the three shape invariants fixed in advance — the positive hump, the first post-hump zero, and the trough — agree with the conjectured density $D(u)$ within the a-priori tolerance $\tau = 0.06$. The invariants are stable across the ladder: the binned hump holds at $u = 0.4625$ and the binned trough at $u = 0.8875$ at every rung, and the first zero stays near $u \approx 0.671$. All values are from the committed `data/m5/ss_x{65536,131072,262144}.txt` runs and the M4 run `data/m4/ss_empirical.txt`:

X	|fam|	hump u	first zero u	trough u	trough depth
10^4	1048	0.4625	0.672894	0.8875	−3.47
2^{16}	5042	0.4625	0.670328	0.8875	−3.72
2^{17}	9014	0.4625	0.673202	0.8875	−3.65
2^{18}	15936	0.4625	0.671945	0.8875	−3.58

(Fig. 1 overlays all four empirical rungs on the conjectured $D(u)$ — each rung a distinct marker — and is regenerated from these committed runs by `figures/make_figures.py`.)

Both estimators, at the verdict rung. The bin width $\Delta u = 0.025$ sets the resolution of the binned argmin/argmax. A parabolic interpolation through the extremal bin and its two neighbours (linear for the zero crossing) gives a sub-bin estimate, and a bootstrap over the curves — the exchangeable unit, $N_{\text{boot}} = 2000$, seed 20260718 committed with the study — gives a 95% confidence interval. At the top rung $X = 2^{18}$, per invariant, against the corrected targets:

invariant	target	binned u	interpolated u	95% CI	deviation
hump	0.465	0.4625	0.47065	[0.46699, 0.47696]	+0.0056
first zero	0.671	0.671945	0.67194	[0.66567, 0.67715]	+0.0009
trough	0.870	0.8875	0.88214	[0.87611, 0.88565]	+0.0121

The deviation is the interpolated position minus the target; the residual statements below ride the interpolated column.

Uniform reading. All three invariants agree with $D(u)$ within $\tau = 0.06$. The first zero sits on target: its CI [0.666, 0.677] contains 0.671. The hump and trough carry small resolved residuals — ≈ 0.006 and ≈ 0.012 — whose confidence intervals exclude their targets marginally (hump: lower bound 0.467, just above 0.465) and clearly (trough: lower bound 0.876, above 0.870); these are resolved offsets, not open deviations, and both sit far inside τ. The largest residual, at the trough, is ≈ 0.012, about a fifth of the tolerance.

The pre-registered verdict, superseded. Before the ladder was extended, PR-1 committed a single decision rule for the trough (`dd6beb0`; the 2^{18} Rung-3 clause `4a17ebe` committed *before the* 2^{17} *rung was read*). With $d(X) = |\text{trough}_u(X) - 0.805|$ and $\Delta u = 0.025$, it declared the displacement finite-X if d recovered by $\geq \Delta u$ at both ladder steps and persistent if d stayed flat. The rule was applied verbatim: $d = 0.0825$ at 2^{16}, 2^{17}, and 2^{18}, flat, with no recovery, so it returned H0 (persistent), read at `8f64ba1`. The audit round then found the target it gated on to be corrupt: 0.805 was our mislocated density trough, not [SS25]'s, the eq (2) transcription error of erratum #28 (§6). The verdict is therefore void as pronounced: the rule was committed and followed honestly, but the target it gated on was wrong. We do not re-run it against the corrected target 0.870, a number it was never registered against. The corrected comparison above is descriptive, and claims no persistence verdict.

Supporting observation (not a gate). The trough *depth* eases across the three extension rungs ($-3.72 \to -3.65 \to -3.58$ at $2^{16} \to 2^{17} \to 2^{18}$) while its position holds; the 10^4 depth (-3.47) is shallower than 2^{16}, so this is a three-rung easing, not a monotone four-rung trend. It has the sign a large-X amplitude decay would carry, but the bootstrap depth intervals overlap heavily across the three rungs, so the easing is a point-estimate trend within sampling noise, not a bootstrap-resolved one. We log it as an observation.

4. Mechanism constraints

Two pre-registered follow-ups were designed to probe candidate mechanisms for the trough deficit: its *position* (PR-2) and the *amplitude* departure at the trough bin (PR-3). After the eq (2) correction the position residual is the small resolved offset of §3, and the amplitude departure is itself modest: the empirical $S(u^*) = -3.58$ against the corrected $D(u^*) = -2.94$ at 2^{18} is a departure ≈ -0.64 in density units, of the same order as [SS25]'s own per-bin discrepancy at this height (§6). Both follow-ups stand as bounds; neither subpopulation accounts for the residual.

PR-2: analytic-rank split (measurement stands; coded verdict superseded). Pre-registered `f7415a4` (the contrast threshold fixed before any curve was split) $\to$ read `876999f`. Excising the 738 analytic-rank-2 curves leaves the trough *position* unchanged — the leave-out trough sits at $u = 0.8875$, identical to the full family, and holds on the geometric-holdout conductor annulus $(10^4, 2^{16}]$; the position is not sensitive to the rank-2 subpopulation. That measurement stands. The pre-committed *recovery gate* that read it, however, asked whether the trough recovered toward 0.805: it gated on the corrupt target, so its verdict is superseded with the rest of the 0.805 chain (§3, erratum #28), while the leave-out measurement it was computed from is untouched. A secondary contrast — the rank-2 curves are markedly more negative on the descending branch (mean gap -1.73, past the committed threshold -0.668) — is a real value-space effect, significant but too dilute to move the position; retained as a measurement. *Wachs clause (verbatim):* this design does not distinguish rank per se from BSD invariants (Tamagawa product, |Ш|, real period) correlated with rank; "carried by" is a statement about the subpopulation, not the mechanism.

PR-3: root-number-imbalance leakage (bounded out). Pre-registered `21060a0` $\to$ read `0d21b62` (the 2^{18} row appended with the Rung-3 run, `8f64ba1`; the leakage table regenerated against the corrected density by `at ss-leakage`). With $\delta = (n_+ - n_-)/|\text{fam}|$ the root-number imbalance, U the unsigned bias, leakage $L = \delta \cdot U$, the trough-bin amplitude departure dep $= S(u^*) - D(u^*)$, and $f = L/\text{dep}$:

X	δ	leakage $L(u^*)$	departure dep(u^*)	$f = L/\text{dep}$	sign vs. departure
10^4	-0.0534	-0.0184	-0.531	$+0.035$	same
2^{16}	-0.0115	$+0.0037$	-0.776	-0.005	opposite
2^{17}	-0.0078	$+0.0025$	-0.713	-0.004	opposite
2^{18}	-0.0103	$+0.0032$	-0.641	-0.005	opposite

The leakage fraction is $\leq 0.5\%$ and opposite-signed (points *against* the departure) at the three reliable rungs, and +3.5% at 10^4 (the smallest family, largest imbalance, no bootstrap); it does not grow with X. Leakage of Sutherland's parity-independent bias through the measured imbalance is bounded out as a material contributor: $f \approx 1$ would require the unsigned bias about two orders of magnitude larger than measured. A citable null; a bound, not an explanation. *Sutherland-bias attribution (verbatim):* the parity-independent unsigned bias, the named congruence classes, and its cancellation in the signed average are Sutherland's (Sept-2023 talk); PR-3 tests only the leakage *relationship*, which is PR-3's hypothesis, not Sutherland's claim.

Congruence stratification: not run. PR-3 fixed in advance the condition under which its more expensive follow-up — a congruence-stratified analysis of the same primes — would be unnecessary: the leakage fraction f small, opposite-signed at the reliable rungs, and with no structured residual at the named congruence classes. The result above meets that condition, so the stratified pass was not run, and no separate pre-registration for it was written.

The small residual of §3 is left unexplained by either constrained mechanism — neither the rank-2 subpopulation (position) nor root-number leakage (amplitude) accounts for it. What could carry it is future work (§6).

5. Reproducibility

A byte-identical published curve. The emitted density JSON is byte-for-byte identical across compilers and platforms. The density evaluator and its emitter (`src/murm/ss_density.cpp`, `src/emit/emit_sawin_sutherland.cpp`) are compiled with fused-multiply-add contraction disabled (`-ffp-contract=off`), so their IEEE operations round the same way under GCC and clang. A freshness check re-emits the snapshot at the current commit and requires byte-equality with the committed file, on CI's GCC as well as a local build. The comparison first normalizes one field — the `generated_by` provenance stamp, a `git describe` string that legitimately varies by checkout — and then requires every remaining byte to match exactly; "byte-identical" below means identical after that single-field normalization.

Integer a_p is platform-independent. A pre-registered, tail-weighted sample of 79,268 (curve, prime) pairs, with primes up to about 4.15 million, was computed by the frozen referee `ap_charsum` on the laptop (`g++-16`, macOS) and on the compute box (`clang 21.1.8`, FreeBSD 15.1). The two outputs are byte-identical, with zero mismatches. This sample corroborates on the 2^{17} family; the primary evidence that a_p is platform-independent is the exact agreement of the three a_p algorithms over the full 2^{16} and 2^{17} grids together with the byte-identical emit above.

The raw partials are the deliberate exception (the contrast). Contraction is off for the density path but left at the compiler default for the empirical partial-sum generator (`src/murm/ss_empirical.cpp` is outside the `-ffp-contract=off` list). So the raw per-curve partial sums drift across toolchains while the emitted density does not. Measured on the shared 8640-curve 2^{17} set, the drift has maximum size 1.887×10^{-15}, the last bit of a value near one. It does not reach the published curve, and byte-identity of the emitted density is established empirically by the freshness check above, not by rounding alone. The precision gap explains it: the emitter writes each value to twelve significant figures (`std::setprecision(12)`, `src/emit/emit_sawin_sutherland.cpp:29`), roughly three orders of magnitude coarser than the drift at the density's order-one scale, so a difference this small falls below the last written figure except at a rounding boundary, which the freshness check would catch.

Errata and inputs. Corrections are logged as numbered errata entries in the research spec and the pinning notes, not edited away. One convention correction: [SS25] eq (1) sums over the half-open prime interval $(C_1N, C_2N]$, and our binning had used the opposite (left-closed) convention (erratum #30). The only pairs it touches are the two-to-three conductor-40 curves per rung whose good primes fall on a bin's right edge, all at low u; no number this note reports changes. The committed runs were regenerated on the compute box under the corrected convention (erratum #30, executed pre-release), so they match the code that produced them, and the change is confined to those enumerated low-u bins. With that regeneration all four canonical rungs are now box-generated (FreeBSD/`clang`); the superseded laptop-origin 2^{16} partials remain in git history, and their line-by-line diff against the current files is an in-repo instance of the cross-toolchain drift above, every difference within the 1.887×10^{-15} bound. Everything the note's numbers depend on is committed, including the from-scratch code, the oracle N/ϵ caches, the per-curve partials, and the `at ap-sample` subcommand. The one input that cannot be regenerated byte-for-byte is the abandoned laptop 2^{17} partials checkpoint, since it is itself the float-drift artifact; it is force-committed through a scoped `.gitignore` exception (`data/m5/ss_partials_x131072.txt.ckpt`), not left to regeneration. The large a_p caches (regenerable by `at ap-cache`) and the cross-platform sample outputs (regenerable by `at ap-sample`) are gitignored and pinned by SHA-256 in the reproducibility spec (`docs/notes/libm-partial-diff-spec.md`).

How to cite and verify. *(Release tag and DOI are placeholders, minted after the author's full review.)* Repository: https://github.com/DerekMarshall/primeknots, release tag `[PLACEHOLDER: v1.0.0]`, at its tagged commit. Zenodo DOI `[PLACEHOLDER]`. To reproduce the results this note reports, from the tagged commit:

```
cmake -B build -DCMAKE_BUILD_TYPE=Release && cmake --build build -j
ctest --test-dir build -L m5 -LE heavy               # M5 suites, ~7 min on a laptop (excludes
 ↪  the heavy 2^18 gate)
./build/bin/at emit --stage m5      --out viz/data/   # empirical-vs-D(u) overlay JSON
./build/bin/at emit --stage m5split --out viz/data/   # rank-split overlay JSON
python3 -m pip install matplotlib                     # figure prerequisite (use a venv; not
 ↪  needed for the suites)
python3 docs/notes/data-note/figures/make_figures.py  # Fig. 1 and the rank-split figure
```

The heavy gate, run separately. The `-LE heavy` above excludes one long check, `m5gate.twin_m0b_bruteforce_x18_tailweighted`, which recomputes on the order of 2.8×10^7 a_p to twin the 2^{18} tail-weighted sample against the frozen referee. It runs about one to two hours on the 48-core compute box (not on a laptop, not in CI) and is invoked explicitly with `ctest --test-dir build -L heavy`.

Regenerating the inputs, and the robustness grid. The large a_p caches regenerate with `at ap-cache --X <H> --cache <ne_cache> --out <bin> --checkpoint <ckpt>` (hours on the compute box for $2^{17}/2^{18}$; SHA-256 and byte counts pinned in `docs/notes/libm-partial-diff-spec.md`), and the run/partials files with `at ss-run`. The referee-round robustness studies are runnable from the committed data with no recompute: the density truncation scan (`at ss-density-scan`, seconds per cutoff, about nine seconds at $B = 2000$) and the bootstrap/interpolation study (`python3 docs/notes/referee_b2b3.py`, under a second), both documented in `docs/notes/referee-round-2026-07.md`.

The cross-platform `a_p` sample is reproduced by `./build/bin/at ap-sample --X 131072 --cache data/m5/ne_cache_x131072.txt`; a single-platform run suffices to verify the committed sample against its pinned SHA-256, and it is the byte-identity of two such single-platform runs (laptop and compute box) that supplies the cross-platform evidence. The pin lives in the reproducibility spec.

6. Discussion

The corrected comparison in context. Agreement with the conjectured density at the heights we can reach is not automatic even for [SS25] themselves. Their Table 4 reports the mean per-bin discrepancy |empirical − density| of their own curve as 1.17, 0.91, 0.73 at 2^{16}, 2^{17}, 2^{18}, decaying to ≈ 0.10 only by 2^{28}, twelve doublings above our top rung. Our rungs sit at the bottom of that window, where the finite-X curve is still far from the limit in amplitude. At these heights the three shape-invariant positions land on the corrected density within τ (§3), while the trough-bin amplitude departure we measure (≈ 0.64 at 2^{18}, §4) is of the same order as [SS25]'s own per-bin figure. We read this as positional agreement with an amplitude that has not yet converged, not as a discrepancy of the replication.

The eq (2) transcription error (erratum #28). An earlier version of this note reported a persistent trough displacement of ≈ 0.0825 against a density target of 0.805; that target was our error, not the data's. Our reading of eq (2) had transposed the two local-factor products (placing $\hat{\ell}$ on the primes $p \mid m$, $p \nmid q$ and ℓ on $p \mid q$) and mis-set the $p \mid q$ exponent to $v_p(m)$ rather than $2v_p(m)$, moving the density trough from its true position near 0.87 to a spurious 0.805. The error was shared. It lived in a pinning note, in the C++ density evaluator built from that note, and in an independent PARI/GP oracle built from the same quote: three internally consistent implementations encoding one reading, so their mutual agreement certified nothing. What caught it was external: the authors' published density figure troughs near 0.87, and a blind re-transcription from the paper against that figure recovered the correct products. The correction and its propagation across the code, data, and prose are erratum #28 and the reframe registry it indexes. The standing rule it produced is that a transcription check terminates at the source artifact, never an internal quote.

Direction of the residuals. The two non-zero position residuals of §3 are both rightward: the hump at $+0.006$ and the trough at $+0.012$ (interpolated, empirical above target), the first zero essentially on target. A single small horizontal dilation of the empirical curve relative to $D(u)$ would produce exactly this, one deformation rather than two independent offsets; it is the natural finite-height reading and sharpens the one follow-up that remains.

PR-4, retired. An earlier draft named a pre-registered split by BSD invariant at fixed rank (PR-4) as the route to whatever carried the trough deficit; the invariants in view (the Tamagawa product, the analytic order of Ш, the real period) are ones Wachs shows modulate murmuration profiles at fixed rank. With the deficit resolved to the small residual above, that motivation is gone, and PR-4 is retired.

PR-5, re-motivated. The live question is whether the coherent rightward shift decays as X grows; the amplitude easing of §3 and the positional dilation here are both consistent with the expected finite-height convergence. PR-5 is a decision rule on that trajectory (position and depth together), to be committed before any rung is added beyond 2^{18}. It is the pre-registered gate for going further, and has not been run.

A scale caveat with a documented precedent. Our computable range, $X \leq 2^{18}$, is small and finite. Sutherland's September 2023 talk gives a precedent for reading such a range cautiously. One formulation of BSD makes the Mestre–Nagao sum $\lim_{x\to\infty} \frac{1}{\log x} \sum_{p\leq x} a_p(E) \frac{\log p}{p} = -r + \frac{1}{2}$ (conditional: Kim–Murty 2023 show that if this limit exists, it equals that value and the Riemann hypothesis holds for $L(E,s)$), and curves of large rank necessarily have large conductor. In Sutherland's rank-stratified plot of that sum, the ordering by rank emerges only at very large conductor. A picture that looks settled at the conductors we can reach need not be the asymptotic one.

Register, restated once. The claim of this note is replication and constraint over the finite range $X \leq 2^{18}$: the three shape invariants agree with the conjectured density within τ, with a small resolved residual (≈ 0.012, largest at the trough), and two candidate mechanisms are bounded out. It is not a claim about the $X \to \infty$ limit [SS25] describe, and nothing in this note bears on that limit or proves anything.
